

Sets

What is a Set



- A **set** is an unordered collection of objects
--> This is not a rigorous definition!!!
- A **definition** reduces a concept to other previously defined concepts
--> "A **COW** is a big animal with horns and four legs in the corners"
- To create mathematical definitions, need some "initial concept" which forms a **foundation** for other definitions. This is usually a "set"
- **Naïve set theory** describes, informally, how sets are built and used
- **Axiomatic set theory (ZFC)** describes a formal, axiomatic system of reasoning about sets

Elements, Describing a Set

- The objects in a set are called **elements** or **members** of the set

$a \in A$ – a is an element of A , a belongs to A

$a \notin A$ – a is not an element of A , a does not belong to A

- One way to describe a set is to list its elements

$\{0,1,2,3,4,5,6,7,8,9\}$ – the set of digits

$\{a,b,c,\dots,x,y,z\}$ – set of Latin letters, alphabet



Extensionality: a set is **defined** by its elements

- A set can be an element of another set

Set of alphabets: $\{\{a,b,c,\dots\}, \{\alpha,\beta,\gamma,\dots\}, \dots\}$

Paradoxes in set theory

- *We should be careful allowing sets to be contained in other sets*
- *Let U be the set of all sets that do not contain itself as an element:*

Strict barbers...?

- *Does U belong to itself?*

If yes, then $U \in U$, hence U does not satisfy the condition, and therefore $U \notin U$.

If not, then $U \notin U$, hence U does satisfy the condition, and therefore $U \in U$.

Universes and classes

- *To avoid paradoxes, separate **sets** and **(proper) classes***
- *Classes are essentially sets --- don't worry about it too much*
- *A **universe** is the class which contains all objects we care about in a given context*
- *Examples:*
 - Set** = *the (proper) class of all sets*
 - Prop** = *the class of propositions*
 - Num** = *the class of numerical sets*
- *Modern foundations treat sets as **concrete objects**, and "classes" as "informal/naive sets"*

But we won't really care about that...

Building sets

● Sets can be described using *set builder notation*:

$\{x \mid P(x)\}$, the set of all x such that $P(x)$

$\{x \mid \text{there is } y \text{ such that } x = 2y\}$, the set of even numbers

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$\{x \mid \exists y (x = 2y)\}$

$\{x \mid x \text{ is a black cow}\}$

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$, the set of natural numbers

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$, the set of integers

$\mathbb{Q} = \{p/q \mid p, q \text{ are integers and } q \neq 0\}$, the set of rationals

$\mathbb{Z}^+, \mathbb{Q}^+$, the sets of positive integers and positive rationals

$\mathbb{R}$, the set of real numbers

Ambiguities in set builder

$\{x \mid 1 \leq x \leq 10\} \leftarrow$ What is this set?

- To avoid such ambiguities, can specify a set of objects from which x is drawn, i.e. a **domain**:

$\{x \in \mathbb{Z} \mid 1 \leq x \leq 10\}$ the set of all integers from 1 to 10

$$= \{x \mid x \in \mathbb{Z} \wedge 1 \leq x \leq 10\}$$

$\{x \in \mathbb{Q} \mid 1 \leq x \leq 10\}$ the set of all rationals from 1 to 10

$$= \{x \mid x \in \mathbb{Q} \wedge 1 \leq x \leq 10\}$$

Equality of Sets, Subsets

- Two sets are **equal** if they have the same elements

That is, $A = B$ if and only if $\forall x (x \in A \leftrightarrow x \in B)$

$$\{1, 3, 5\} = \{5, 3, 1\}$$

$$\{1, 3, 5\} = \{5, 5, 5, 5, 3, 3, 1, 1, 3, 1\}$$

$$\{1, 3, 5\} \neq \{1, 3, \{5\}\}$$



- Set B is a **subset** of a set A if every element of B is also an element of A .

That is $B \subseteq A$ if and only if $\forall x (x \in B \rightarrow x \in A)$

Example:

$\{x \in \mathbb{Z} \mid 1 \leq x \leq 10\}$ describes a subset of $\mathbb{Z}$

More definitions

● Set B is superset of A , denoted $B \supseteq A$, if A is a subset of B

● Set B **is not** a subset, denoted $B \not\subseteq A$ of a set A if there exists an element in B but not A

In symbols: $B \not\subseteq A \Leftrightarrow \exists x (x \in B \wedge x \notin A)$

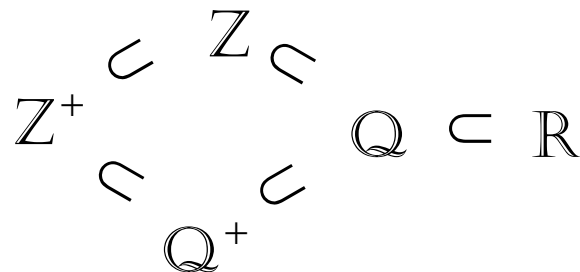
● A set B is said to be a **proper subset** of a set A , denoted by $B \subset A$, if it is a subset of A and is not equal to A .

In symbols: $B \subset A \Leftrightarrow (B \subseteq A) \wedge \exists x (x \in A \wedge x \notin B)$

$$\{1,3\} \subset \{1,3,5\}$$

$$\{1,3,5\} \supseteq \{1,3\}$$

$$\{2,6\} \not\subseteq \{1,4\}$$



Transitivity of subsets

● **Theorem.** *If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.*

Proof.

We have to prove that, for every x , if $x \in A$, then $x \in C$.

Take any x such that $x \in A$. Since $A \subseteq B$, this implies that $x \in B$.

Next, as $B \subseteq C$ and $x \in B$, we have $x \in C$.

Q.E.D.

Homework: give a formal proof, using rules of inference

Equality and subsets

● **Theorem.** $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Proof.

If direction:

Suppose $A = B$.

Then for every x , if $x \in A$, then $x \in B$, so $A \subseteq B$

Likewise with $B \subseteq A$

Only if direction:

Suppose $A \subseteq B$ and $B \subseteq A$

Then if $x \in A$, then $x \in B$ by $A \subseteq B$

And if $x \in B$, then $x \in A$ by $B \subseteq A$, so $A = B$

Q.E.D.

Empty Set

● *Empty set is a set that has no elements.* $\emptyset$

How many elements does this set contain? $\{\emptyset\}$

● **Theorem.** *For any set A , (i) $\emptyset \subseteq A$, and (ii) $A \subseteq A$.*

Proof.

(i) We have to prove that $\forall x (x \in \emptyset \rightarrow x \in A)$

Since the premise $x \in \emptyset$ is always false, we conclude that the implication $x \in \emptyset \rightarrow x \in A$ is always true.

Then we use the rule of universal generalization.

(ii) Homework.

Cardinality

- Let A be a set. If there are exactly n distinct elements in A where n is a natural number, then we say that A is a **finite set** and that n is the **cardinality** of A .

$$|A| = n$$

- Examples

$$|\emptyset| = 0$$

$$|\{1, 2, 3, 3\}| = 3$$

$$|\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}| = 10$$

$$|\{a, b, c, \dots, x, y, z\}| = 26$$

- Sets that are not finite are called **infinite**

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are infinite

Power Set

- Given a set A , the **power set** of A is the set of all subsets of A

$$P(A) = \{ B \mid B \subseteq A \}$$

- Example.* Find $P(\{1,2,3\})$

- Theorem.** If A is a finite set, then $|P(A)| = 2^{|A|}$

Proof (sketch)

Let B be a subset of A and $|A|=n$.

We can describe B by which of the n elements of A it contains

Let $i_1i_2i_3\dots i_n$ be the binary string where $i_j = 1$ if and only if B contains the j th element of A

How many unique strings (and hence subsets B) are there?

Practice

Exercises from the Book:

7th edition: 1, 5, 7, 9, 23 (page 125 – 126)

8th edition: 1, 7, 9, 11, 25 (page 131 – 132)